

Americas Energy & Transition

Americas Power & Energy Transition: Paying for speed vs. scale - The BTM/FTM/BYOG Framework for BE, FRVO, and ORA's Hyperscaler Deals



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- BE, FRVO, and ORA now all have live, named hyperscaler or utility-anchored deals, but they sit in three structurally different boxes: Bloom is true behind-the-meter (BTM, zero interconnection exposure, ~90-day deployment); Ormat and most of Fervo's contracted volume are front-of-meter (FTM, grid-delivered, still queue-exposed); and Bloom's AEP deal is a third, hybrid category — utility-financed onsite generation ("BYOG") that could scale further than either pure model.
- Bloom wins on speed and optionality — 55- 90 days vs. 3–8 years for standard interconnection — but pays for it: illustrative LCOE of \$85–115/MWh driven by natural gas fuel exposure and a recurring stack-replacement cost.
- Fervo's economics are a bet on its learning curve: Cape Station 1 wells were at ~\$7,000/kW, with the holy grail is hitting its stated \$3,000/kW target — which would make FTM geothermal cost-competitive with BTM fuel cells without the fuel-price exposure.
- Ormat is the smallest, lowest-risk mover — brownfield uprate-style economics on existing geothermal assets — but its NV Energy Clean Transition Tariff deals are capped by resource availability (13–150 MW), not technology cost, and we don't think will scale to gigawatt volumes the way Bloom or Fervo can.

As PJM, ERCOT, and FERC push interconnection reform (fast-track cycles, Order 2023, Batch Zero), the queue-bypass premium BTM currently commands should compress over the back half of this decade.

BERNSTEIN TICKER TABLE

Ticker	Rating	10 Jul 2026		Price Target	TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
		Cur	Closing Price			Cur	2025A	2026E	2027E	2025A	2026E	2027E
BE (Bloom Energy)	M	USD	244.61	276.00	843.0%	USD	0.82	2.30	4.27	296.9	106.2	57.3
FRVO (Fervo)	O	USD	27.13	47.00	NA	USD	(0.25)	(0.10)	(0.16)	(109.2)	(277.2)	(172.3)
ORA (Ormat Technologies)	U	USD	109.77	115.00	4.9%	USD	2.02	2.26	2.48	54.4	48.5	44.2
SPX			7,515.34									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

BE estimate is Adjusted EPS; BE valuation is Adjusted P/E (x);

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We are Market-Perform on Bloom, Outperform on Fervo, and Underperform on Ormat

DETAILS

1. THE FRAMEWORK

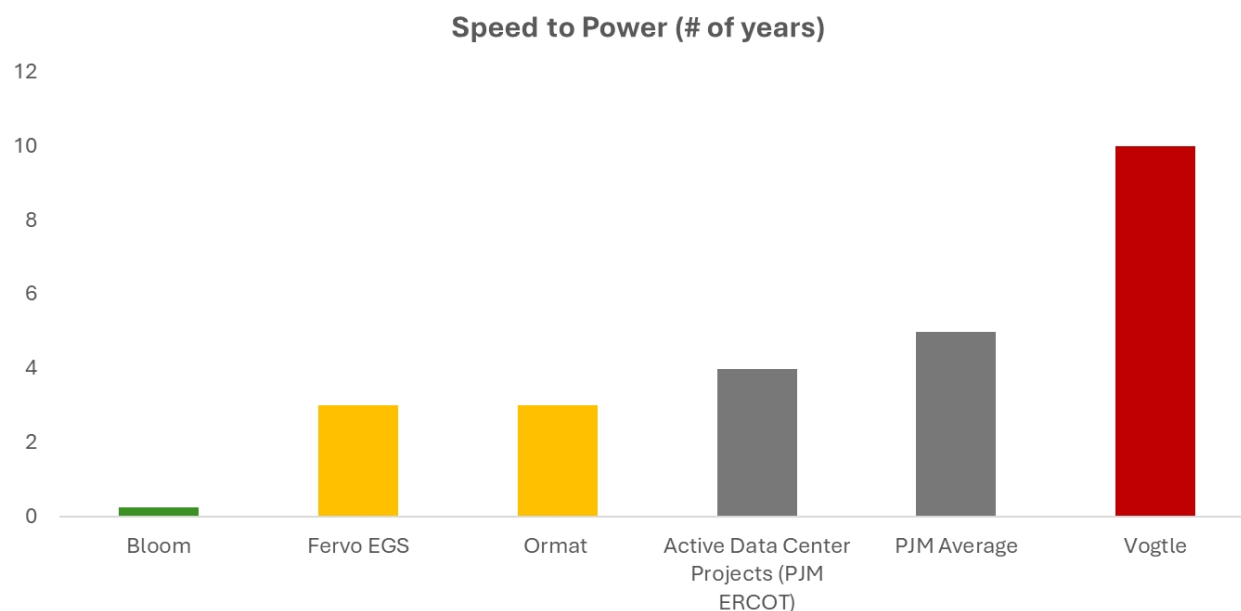
Three structurally distinct models are now competing for the same hyperscaler capital, and the distinctions matter more than the shared "data center power" label suggests.

EXHIBIT 1: **Different models of powering up**

Model	Definition	Interconnection exposure	Representative deal	Typical speed
BTM (behind-the-meter)	Generation sited and consumed on the customer's side of the utility meter; the utility interconnection process for that load is bypassed entirely.	None for the covered load — no generation or large-load interconnection queue involved.	Bloom/Oracle Project Jupiter — up to 2.8 GW MSA, full onsite power for the New Mexico AI campus.	~55-90 days from order to energized (Bloom's stated figure).
FTM (front-of-meter)	Generation interconnects to the grid and sells through a utility or ISO; the customer's load is served via a PPA sleeve, not a direct wire.	Full generation-interconnection queue exposure, plus transmission/delivery risk to the load.	Ormat/Switch Salt Wells (13 MW); Ormat/Google NV Energy portfolio (150 MW); most of Fervo's SCE and NV Energy contracted volume.	2–4+ years from contract to energized power, even under expedited tariff structures.
BYOG (hybrid) ("bring your own generation")	Utility contracts for and finances dedicated onsite/near-site generation to serve a large load its own grid capacity cannot accommodate — capital sits on the utility's books, physically resembles BTM.	Utility-managed; avoids the large-load queue but the utility itself absorbs project and fuel-supply risk.	Bloom/AEP — \$2.65B, up to 1 GW, 20-year offtake, announced Jan. 2026.	Faster than standard interconnection; slower than pure BTM given utility procurement processes.

Source: Bernstein Analysis, Company filings

BTM revenue is largely insulated from the interconnection crisis that is otherwise the sector's central bottleneck (see prior notes on PJM capacity auctions and the interconnection queue). FTM revenue is not — it is simply competing for the same constrained queue slots as everyone else, with a corporate offtaker's name attached. BYOG is the wildcard: it turns the utility itself into the entity racing the queue on the customer's behalf, which is a different risk profile (utility credit, utility execution) than either pure model.

EXHIBIT 2: **Speed to Power: BTM vs. FTM, vs. the last standard nuclear plant built in the US for reference**

Source: Bernstein Analysis, Company filings, Deal announcements for Bloom

2. WHERE BE, FRVO, AND ORA SIT

Bloom Energy – BTM plus a new BYOG wrinkle

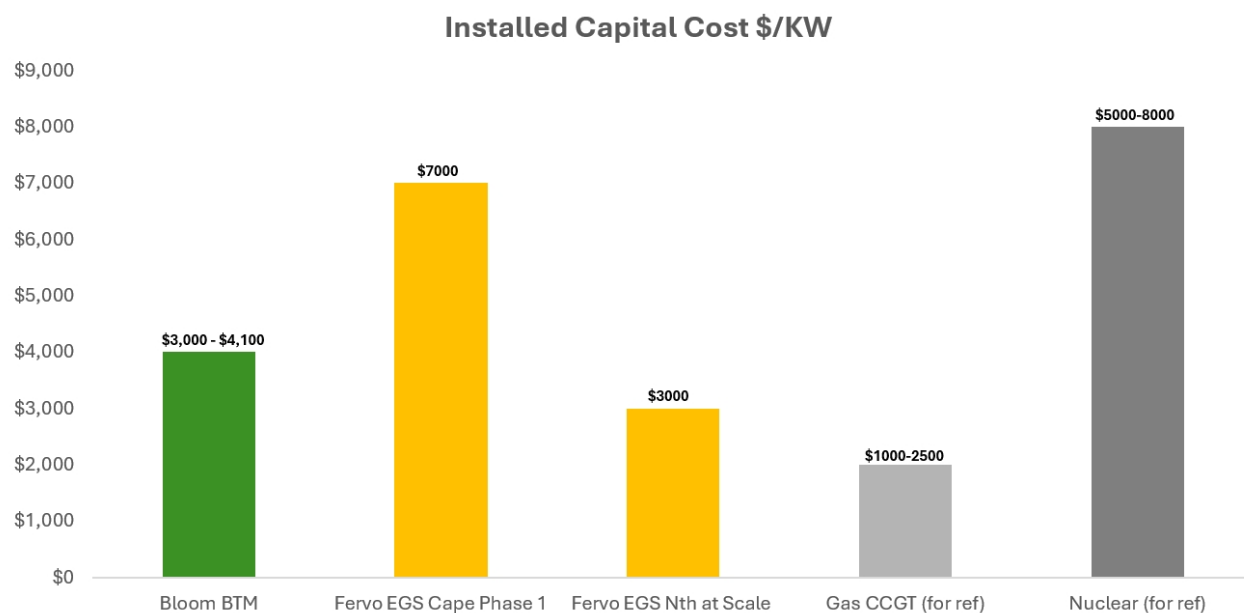
Bloom's ~2.8 GW Oracle master services agreement (April 2026, 1.2 GW already contracted for 2026–2027 delivery) is textbook BTM: fuel cells sited at Oracle's Project Jupiter campus, DC-coupled via Bloom's native 800V DC architecture directly into IT racks, bypassing both the AC/DC conversion stack and the grid interconnection process for that load. The January 2026 AEP agreement (\$2.65B, up to 1 GW, 20-year offtake) is a different animal: AEP is contracting for Bloom capacity to serve large loads its own grid cannot accommodate quickly enough. The generation is still physically onsite/near-site, but the offtake counterparty is a utility, not the end customer — a hybrid structure with the utility bearing the credit and financing risk rather than the hyperscaler.

Fervo Energy – FTM at scale, betting on the cost curve

Fervo's March 2026 Geothermal Framework Agreement with Google — up to 3 GW through 2033, with 1 GW prioritized in the first two years — is FTM: power is delivered through utility intermediaries (NV Energy, Southern California Edison) under conventional PPA structures, meaning every megawatt is still subject to generation-interconnection review even where NV Energy's Clean Transition Tariff is designed to streamline the commercial and regulatory path. Fervo's IPO filing discloses Cape Station is being built today at roughly \$7,000/kW — competitive with nuclear, but more than three times the cost of a gas plant — with a stated target of cutting that to \$3,000/kW through repeatable drilling and pad design.

Ormat – the smallest, most de-risked mover

Ormat's two 2026 data center deals — the 13 MW Switch PPA at its existing Salt Wells plant (January 2026) and the up to 150 MW Google/NV Energy portfolio PPA (February 2026) — are also FTM, and notably conservative: Salt Wells is a brownfield upgrade to an already-interconnected facility, the geothermal equivalent of CEG's Braidwood/Byron uprate playbook rather than a greenfield EGS bet. Both deals run through NV Energy's Clean Transition Tariff and await PUCN approval (expected second half of 2026), with the Google portfolio not online until 2028–2030. Ormat is monetizing existing, de-risked assets at small scale.

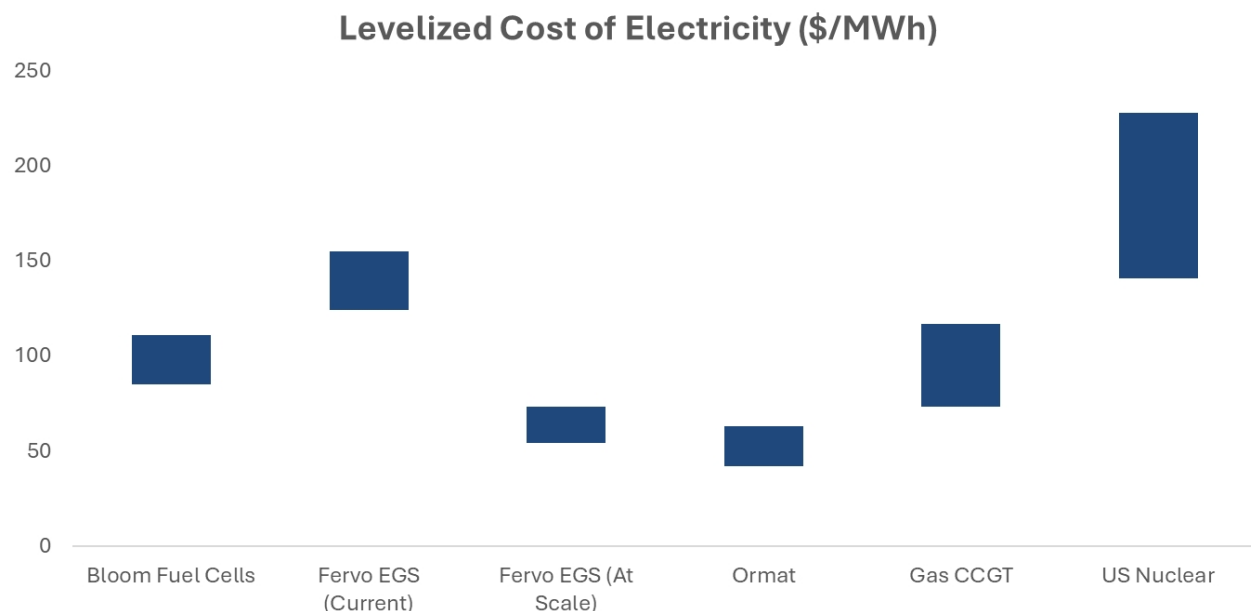
EXHIBIT 3: **Installed Capex - Fuel cells vs. EGS vs. gas or nuclear for reference**

Source: Bernstein Analysis, Fervo S1 filing, Lazard LCOE 2025

3. ILLUSTRATIVE ECONOMICS: PAYING FOR SPEED VS. PAYING FOR SCALE

Bloom's installed cost runs \$3,100–4,000/kW pre-incentive, falling to roughly \$2,200–2,800/kW after the restored Section 48E investment tax credit (30%, technology-neutral, in effect through 2033). Layering in natural gas fuel cost (running ~10% above a combined-cycle plant given Bloom's ~54% electrical efficiency, close to CCGT heat rates) and a recurring stack-replacement reserve (roughly \$1,000/kW every five - seven years, about 30% of system cost), an illustrative LCOE lands in the \$82–115/MWh range — pre- vs. post-ITC. That is a real premium to grid wholesale power in most markets, and Bloom's own pitch is explicit that the value proposition is speed and reliability (up to five-nines availability), not lowest-cost energy.

Fervo's math is the mirror image. At today's ~\$7,000/kW, Cape Station's illustrative LCOE is comparable to Bloom's — roughly \$92–112/MWh — but with no fuel-price exposure and a resource that doesn't degrade. If Fervo hits its stated \$3,000/kW target, that LCOE compresses to roughly \$48–60/MWh, cheaper than Bloom's post-ITC case and competitive with new gas. Cape Station Phase I is Fervo's first commercial-scale project, and the company has to demonstrate the learning curve. Ormat's brownfield economics are directionally the cheapest of the three (existing infrastructure, no new wells), but the model doesn't scale the way Fervo's repeatable GeoBlock design is designed to.

EXHIBIT 4: **Directional LCOE for Fuel Cells and EGS**

Source: Bernstein Analysis, Lazard LCOE 2025

4. WHO ACTUALLY WINS THE QUEUE-BYPASS RACE?

Short-run (2026–2028 delivery): Bloom. The gap between a 90-day BTM deployment and a 3–4-year FTM interconnection wait for even the most privileged data center projects in PJM and ERCOT is the whole story. Hyperscalers with near-term capacity gaps — Oracle's Project Jupiter is the clearest example — have no substitute for BTM at this horizon, and AEP's willingness to underwrite Bloom capacity as a utility-financed BYOG asset suggests utilities themselves are starting to view onsite generation as a capacity-planning tool rather than a customer workaround.

Medium-run (2028–2033 delivery): If Fervo can demonstrate EGS at scale at \$3,000/KW, FTM geothermal becomes both cheaper than BTM fuel cells and free of fuel-price risk, which should pull incremental hyperscaler capital toward Fervo on any deal with a delivery date past 2028 — provided the interconnection queue for that specific project clears on schedule, which is the load-bearing assumption Google's Nevada 115 MW deal (targeting 2030) and the broader Google GFA are making. Ormat's brownfield model wins on risk-adjusted terms but is structurally capped by how many uprate-able geothermal assets actually exist near data center clusters — in our view, it is a scarce, not a scalable, source of supply.

Structural risk to the entire BTM premium: interconnection reform is a real, funded policy priority — PJM's fast-track cycles, FERC Order 2023 cluster studies, and ERCOT's Batch Zero process are all explicitly aimed at cutting queue times. If any of these reforms bite meaningfully before 2030, the speed premium that justifies Bloom's \$82–115/MWh over FTM alternatives narrows, and premium should flow toward whichever FTM name has the better underlying cost curve — which, on current disclosures, points to Fervo if it executes.

5. RISK AND CATALYSTS

Risk	Detail	Why it matters
Bloom's gas-price and 45V exposure	BTM LCOE is directly geared to natural gas prices; hydrogen fuel-flexibility optionality depends on 45V credit economics that <u>remain</u> a smaller, less certain lever than the 48E ITC.	A sustained rise in gas prices would erode Bloom's cost advantage relative to grid power in exactly the markets where <u>hyperscalers</u> have the most FTM alternatives.
<u>Fervo's</u> journey ahead	Cape Station Phase I is <u>Fervo's</u> first commercial-scale project; the \$3,000/kW target is a company target that needs to be achieved when EGS gets to scale	If <u>Fervo's</u> cost curve <u>stall</u> , the <u>medium-run</u> "FTM wins" case for geothermal weakens and BTM's premium continues.
Interconnection reform pace	PJM fast-track, FERC Order 2023, and ERCOT Batch Zero are all designed to cut queue times but are early in implementation.	Faster-than-expected reform compresses BTM's speed premium; slower-than-expected reform extends Bloom's runway and the case for further utility BYOG deals.
48E ITC and tax-credit transferability	Bloom's post-ITC economics depend on continued 48E availability and monetization mechanics (direct pay, transferability) remaining intact.	Any legislative or IRS-guidance change to 48E treatment would move Bloom's illustrative LCOE back toward the pre-incentive end of the range.

BOTTOM LINE

BE, FRVO, and ORA are not really competing in the same race. Bloom is selling speed and is priced accordingly; Ormat is selling low-risk incrementalism on a resource base that we don't believe lends itself to scale (unless they crack the EGS puzzle); Fervo is the only one of the three underwriting a cost-curve bet, and it is the name most exposed to execution risk and most rewarded if that bet pays off. For a coverage universe organized around "who wins the interconnection bypass," the honest answer is that Bloom wins today, Fervo wins the argument if Cape Station's cost curve proves out by 2028, and Ormat wins on risk-adjusted return without ever needing to be the largest name in the room. We'd treat Fervo's Phase I commissioning (Q4 2026 target) and any updated cost disclosure toward its \$3,000/kW target as the single highest-signal near-term catalyst for the FTM side of this framework.

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VALUATION METHODOLOGY**Bloom Energy Corporation**

We reach our target price of \$276/sh by applying a 11x revenue multiple on 2027-28 estimated revenue of \$6.9B.

Fervo Energy Company

EV to EBITDA. We use 2035 Adjusted EBITDA and an EV/EBITDA multiple of 10.5x and discount it back to 2027 for a price target of \$47

Ormat Technologies, Inc.

We use a 14.5x multiple on 2027-28 estimated adjusted EBITDA of \$658MM to arrive at our target price of \$115/sh.

RISKS**Bloom Energy Corporation**

Key downside risks to our price target include grid upgrades limiting demand for Bloom's onsite power solution, and lower than expected data center demand. Higher than expected demand for onsite power and contracting activity are upside risks.

Fervo Energy Company

Downside risks to our price target include: Lack of power demand growth, or weaker than anticipated power prices for future PPAs Competing technologies → Fuel cells, Solar plus battery storage Permitting challenges and changes to overall government support Unexpected operational challenges as phase I project ramps Capital project over runs on future phases Inability to drive down future project costs Heat transfer/ faster than expected cooling

Ormat Technologies, Inc.

Upside risks to our price target include 1) the inability of competitors to prove and scale enhanced geothermal technology; 2) greater than anticipated PPA contracting for behind the meter power with hyperscalers for the long term; 3) the development and commercialization of EGS by Ormat themselves before competitors; and 4) significant upside from storage margins if key electricity markets get further constrained.

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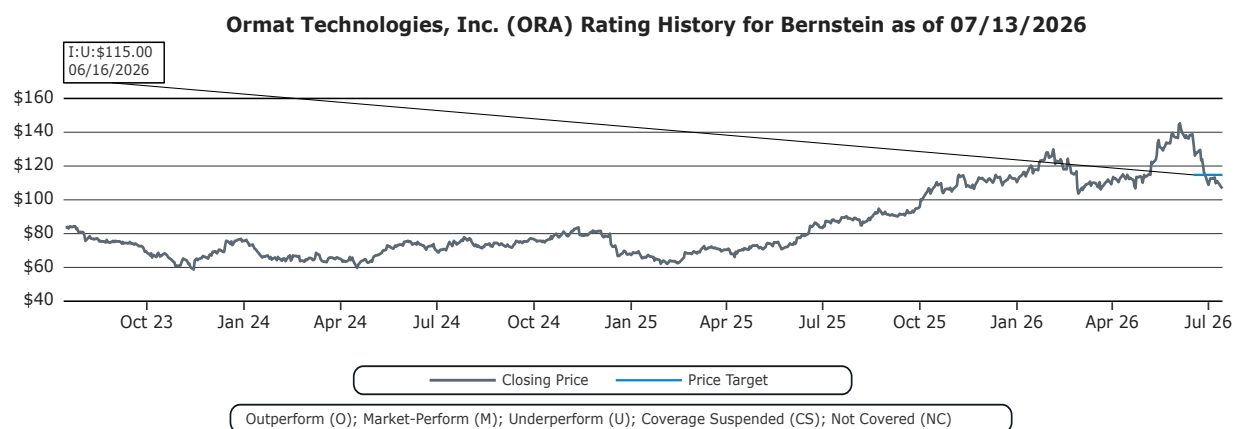
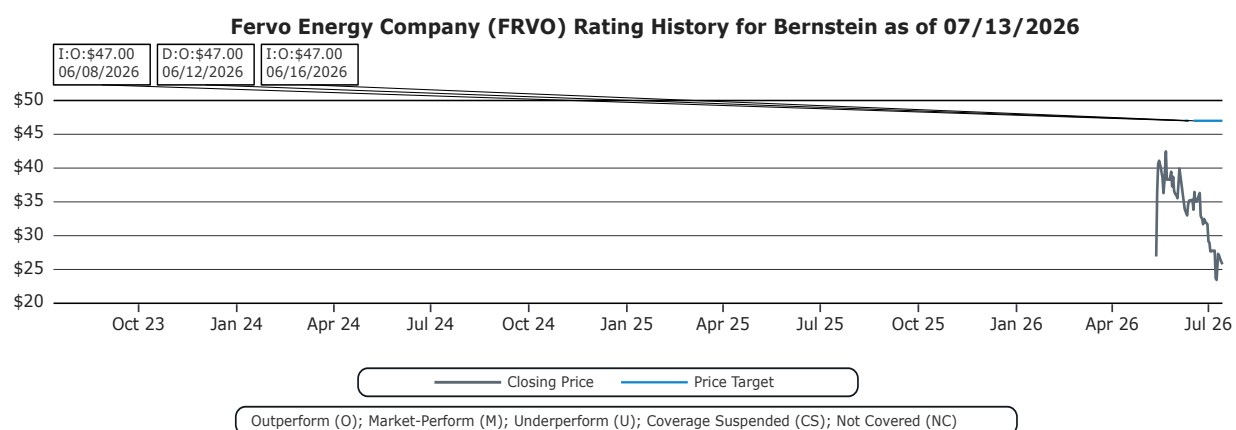
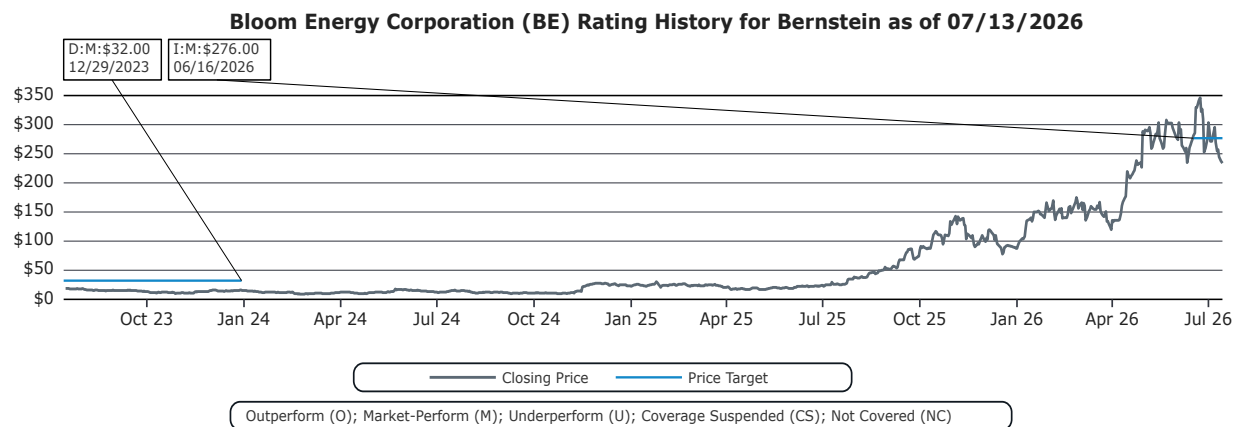
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